Structure-Aware Diversity Pursuit as an AI Safety Strategy against Homogenization

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Abstract

Generative AI models reproduce the biases in the training data and can further amplify them through mode collapse. We refer to the resulting harmful loss of diversity as homogenization. Our position is that homogenization should be a primary concern in AI safety. We introduce *xeno-reproduction* as the strategy that mitigates homogenization. For auto-regressive LLMs, we formalize xeno-reproduction as a structure-aware diversity pursuit. Our contribution is foundational, intended to open an essential line of research and invite collaboration to advance diversity.

1 Introduction

> *But even if we are not here next year, our DMs, our selfies, our late-night voice notes, they'll be. Our memory is the archive now.*
>
> *@bundleof_styx*
>
> *July 28, 2025 on Reels*

In this epigraph, trans intellectual bundleof_styx laments the recent transphobic turn in the United States, a shift that threatens the survival of her community. The stories in the margins have historically been excluded from the archive (Spivak, 1988), so their memory faded with them. Today, however, the internet allows (and forces) the recording of many more stories. These are still very subtle traces against the dominant narratives (Hussain, 2024). **How should technology respond to the faint echoes of the minoritized?**

AI safety recognizes that AI systems can amplify biases leading to concrete harm (Bengio et al., 2025). However, AI safety usually differentiates and prioritizes future catastrophic risk over present social harm (Morozov, 2024; Harding and Kirk-Giannini, 2025). In this paper, we respond to traces from the margins by foregrounding them within AI safety discourse through a focus on diversity.

The purpose of this paper is to foreground **homogenization**, the harmful loss of diversity in outputs (Rudko and Bashirpour Bonab, 2025; Agarwal et al., 2025; Hussain, 2024; Sourati et al., 2025; Moon et al., 2025) from Generative Artifical Intelligence (GenAI). Borrowing terminology from critical theory (Hester, 2018)[1], we call a strategy **xeno-reproductive** if it counteracts homogenization. **Xeno-reproduction** is the generative process that intentionally increases diversity.

Our main standpoint is that diversity is always relative to a context. We take the first steps to operationalize this principle by offering an abstract framework that aims to encapsulate some nuances of context. Our framework can be thought of as **structure-aware**, as it offers a vocabulary of structures, systems, and compliances. Given that an LLM defines a probability distribution over all possible trajectories, we **enhance our structural account with string statistics**. This allows us to further introduce the notions of cores, orientations, deviances, and dynamics. Finally, our formalism enables us to formalize xeno-reproduction.

Our contributions:

- We motivate the formalization of xeno-reproduction as a core AI safety strategy. (Section 2)
- We provide an expressive theoretical framework that allows us to jointly reason about the structures and the statistics of strings. (Section 3)

[1] We adopt critical theory language because technology is outpacing traditional concepts (Hadfield, 2023), and stale language fails to make the impacts of our theorizations explicit. A theory with teeth, one that is attuned to real stakes (Saketopoulou, 2023), must remain groundbound (Bettcher, 2025), foregrounding minoritized people rather than disembodied abstractions. Would it not be naive to try to "solve diversity" without engaging with the academic fields that explicitly study social bias (e.g., Queer Theory, Postcolonial Studies, Black Studies)?

- We formalize homogenization (Section 4) and xeno-reproduction (Section 5).

Our position is that AI safety should center homogenization in its research and mitigation agenda, and that structure-aware diversity pursuit is a key part of the strategy to address homogenization in LLMs. The goal of this paper is not to present a complete and empirically validated algorithm, but rather to offer a conceptual vocabulary and formal scaffolding to guide future research on diversity in LLMs.

2 Background

A case against homogenization is a case for diversity. Roughly, we can think of the **diversity of a community as the average rarity of its members** (Leinster, 2024). For a community of LLM outputs, a string is rare if it is generated infrequently, and similar strings are also generated infrequently. However, people tend to disagree on what kind of similarities and differences are meaningful (Vrijenhoek et al., 2024). Embracing ambiguity (Reinhardt, 2020) for us amounts to attending to context. This section situates diversity in the contexts meaningful to us, guiding our desiderata for xeno-reproduction.

2.1 Why is diversity lost?

The initial driver of diversity loss is the way our data is collected (Guo et al., 2024b). **The archive** refers to the corpora of training data encoding historical pattern, which does not fully or accurately represent reality. Minoritized populations are often underrepresented or **misrepresented** (Bengio et al., 2025).

Even if our training data perfectly reflected the world, generative models (Huang and Huang, 2025) generally do not capture the complete diversity of the training data. This phenomenon has been referred to as **mode collapse** (Jiang et al., 2025), a failure of distributional faithfulness that negatively impacts diversity. It was initially introduced in the context of GANs (Huang and Huang, 2025). For LLMs, the terminology has been somewhat loose (Schaeffer et al., 2025). Generalized mode collapse encompasses mode dropping (Huang et al., 2024; Yazici et al., 2020), no-breadth scenarios (Kalavasis et al., 2025b), coverage collapse (Schaeffer et al., 2025), overgeneralization (Li and Farnia, 2023), mode interpolation (Aithal et al., 2024), degeneration (Finlayson et al., 2023), and catastrophic forgetting (Cobbinah et al., 2025; Thanh-Tung and Tran, 2020).

Mode collapse amplifies the biases already baked in by societal structures. Dominant modes in the data reflect societal biases. When out GenAI models preserve only these dominant modes, the modes corresponding to minoritized populations are erased.

2.2 Why is diversity important?

There are always rare events of interest in the long tails of reality's distribution. For example, we want to understand, model, and prepare for extreme catastrophes (Gu et al., 2025), such as unexpected natural disasters. Similarly, we want to reproduce those rare bursts of genius that generate novel, paradigm-shifting innovations in our research work (Uzzi et al., 2013; Hofstra et al., 2020; Wu et al., 2019). We find examples of interesting rarity in all domains (Stanley and Lehman, 2015), including: web server computing (Dean and Barroso, 2013), market research (Von Hippel, 1989), autonomous vehicles (Putra et al., 2024), cybersecurity (Edwards et al., 2016), and ecology (Leitão et al., 2016). How do we guide our GenAI models to reproduce the realities found in these long tails?

Outliers (Bhandari et al., 2024) and anomalies (Ruef and Birkhead, 2024) are powerful (Beamish and Hasse, 2022; Cook et al., 2021). Each instance represents a possible real mechanism that we have not yet considered (Woodward, 2005; Rudman et al., 2023). Because we lack understanding, they often escape our systems of classification (Bowker and Star, 1999). Even experts can confuse (Sokol and Hüllermeier, 2025) aleatoric and epistemic uncertainty.

Some of the long tails of reality originate from structural inequity in society (Schwartz et al., 2022; Lopez, 2021). Without any intervention, GenAI is expected to worsen the lives of those minoritized (Hussain, 2024). The traces from the minoritized are not only faint but also often overlooked (Jasanoff, 2007; Mohamed et al., 2020) and even actively silenced (McQuillan, 2022). The result is that we do not even know what to look for, even when they are right in front of us (Gopinath, 2005). **Some of the most ethically important long-tail cases will be hard to detect**.

2.3 Who does homogenization harm?

Homogenization harms the minoritized. As social bias is amplified, all its associated

harms are also intensified, including representational (Katzman et al., 2023), allocational (Shelby et al., 2023), and narrative[2] (Coeckelbergh, 2023) harms.

Homogenization also harms everyone else. Against expectations, the margins continue to be powerful sources of creative production for society (Forgeard, 2013; hooks, 1990; Betts et al., 2017; Hamraie and Fritsch, 2019; Tedeschi and Calhoun, 1996; Gould, 2009). The structural violence from oppression never quite stifles it (Galtung, 1969; Farmer, 2004). From critical theory (Wark, 2023; Harney and Moten, 2013; Coleman, 2021), we associate the unique energetic source in the margins with **the surround**: the field beyond what can be surveilled, disciplined, and contained. When homogenization erases the traces of deviance, it does not merely harm the minoritized. It impoverishes everyone.

2.4 What are the existential risks of homogenization?

Narrative and storytelling are some of the oldest and most powerful technologies (Zurn et al., 2024). With phenomena like AI-induced psychosis (Preda, 2025), we are just beginning to grapple with the profound ways that LLMs can shape our minds and behavior. Over time, if LLMs deliver too little diversity (Bommasani et al., 2022), our ability to interpret our own experiences and entertain alternative possibilities will shrink (Gillespie, 2024). Eventually, homogenization leads to future knowledge collapse (Peterson, 2025), degradation of innovation, and erosion of the human experience (Han, 2024; Berardi, 2017; Preciado, 2013).

The last few years have made it clear that even "less advanced" technology, such as social networks, can have enormous negative impacts (Allcott et al., 2020). Algorithmic recommendations can also have a homogenizing effect, as they tend to standardize and narrow discourse (Putri et al., 2024). This fosters echo chambers and filter bubbles that amplify polarization and misinformation (Rodilosso, 2024). Tragically, in some cases, these dynamics have escalated into **real-world violence** (Facebook, 2021) and even genocide (Modok, 2023). This foreshadows the near-term existential risks of AI, especially as it becomes more powerful and more deeply integrated into our lives (Bucknall, 2022; Kasirzadeh, 2025; Kolt, 2024).

2.5 Why is diversity complex?

Diversity is complex (Mironov and Prokhorenkova, 2025) because it is only meaningful in relation to a **context** (Peeperkorn et al., 2025). When measuring diversity, are we concerned with lexical variation or with gender representation? These are very different things. Most existing techniques to increase diversity in LLM outputs overlook this, and often fail in practice. For instance, increasing temperature increases incoherence more than novelty (Peeperkorn et al., 2024), limiting usefulness before hitting text degeneration (Lee et al., 2025). Despite hyperparameter tuning, homogeneity bias persists and is particularly pronounced for minoritized groups (Lee, 2025). Advanced prompting techniques (which have been effective for reasoning tasks) do not help increase creativity in outputs (Morain and Ventura, 2025).

Not only do **we lack reliable ways to increase the diversity of LLM output**, but **current alignment methods are actively reducing it**. Recent literature (Murthy et al., 2025; West and Potts, 2025; Meng et al., 2024) has shown that alignment degrades the capabilities of LLMs related to output diversity. The trade-offs introduced by alignment are only now coming into focus, but there is a clear narrowing of the generative horizon (Yang and Holtzman, 2025; Feng et al., 2025). **How to make AI safe without making it homogenizing remains understudied**.

2.6 Diverse how, anyway?

Recent work challenges the assumption that hallucinations are always problematic or undesirable (Yuan et al., 2025; Sun et al., 2025). Since diversity is task-dependent (Jain et al., 2025), **what counts as a hallucination is rather a prescription**[3].

Indeed, many formalisms (Li et al., 2025) take a normative (Sui et al., 2024) approach to defining hallucinations, such as formulating the binary classification problem "Is it Valid?" (Kalai et al., 2025). However, we recognize that there are many

[2] Narrative harms can also be considered as aspirational (Fazelpour and Magnani, 2025), imaginative (Gillespie, 2024), and epistemic (Barry and Stephenson, 2025) harms, or hermeneutic (Goetze, 2018) injustices.

[3] Appendix B shows that hallucination frameworks and language generation theory can be recast within our vocabulary, suggesting potential for theoretical unification.

ways for a model to hallucinate (Huang et al., 2025; Cossio, 2025), and we advocate for sufficiently expressive formalisms[4] so that rich productive variation is not dismissed as a mere defect. What counts as valid depends on context, and specifying that context requires nuance.

2.7 What insight does Critical Theory provide?

Ahmed (Ahmed, 2006) reminds us that we come into every experience *oriented*: nothing simply exists on its own, but is always facing a certain way, closer to some things and farther from others. **Orientation is the way in which social and cultural norms direct us toward certain objects, people, and life paths, making some information more proximal, accessible, and legible than others.** It determines what is within reach, and what is further away.

Normativity is the set of orientations that power has laid out for us to collectively follow. It is the default path formed by repetition over time, and how historical conditions shaped what feels natural and what aligns with dominant modes of the data. A normative orientation follows this pull from the default while a **queer orientation** deviates from it, reaching toward what the norm occludes. By "queer," we refer not to a specific identity but to a general structural condition: that which moves against the default orientation of a system. **Queerness is the deviance from dominant modes, non-normativity defined with respect to a reference frame. Whatever a system treats as the default path, queerness refers to what steers away from it.**

2.8 What do we want from the future?

To promote diversity, GenAI should:

- **Be queer**: Diverge into the long tails.
- **Center the subaltern**: Attend to traces rendered invisible by power.
- **Be explicit**: Specify what diversity means in context.

3 Theoretical Framework

3.1 The Big Picture

The **context is the determination of what is important in a setting**. We decompose the context into **situation** (locality) and **interface** (subjetivity). In LLMs, the situation is captured by the prompt, but what about the interface? How do we encode our values, prescriptions, and politics? We propose scoring a string to characterize its alignment with our judgments.

Abstractly, we can think of asking whether a given string exhibits a behavior of interest. Does this output follow English syntax? Does it mention women in positions of power? We call any such specification of organization among tokens a **structure**. We use *structure compliance* to refer to the scoring of a string based on that structure. The term is intentionally dual: it evokes the abstract notion of a mathematical pattern while also hinting at structures of power. Multiple structures form a **system**.

The **core** is the characterization of normativity given a system. It describes what the model treats as the default for each structure. The core allows us to define **orientation**, which measures how normative or deviant an output is. **Homogenization** collapses toward the core while **xenoreproduction** diverges from it.

3.2 Measuring Normativity

There are many statistical ways to characterize the default: The mean, the mode, the greedy path. Each characterizes normativity differently. Properly analyzing diversity requires reporting: **context** (prompts and structures/systems) and **core characterization** (the statistics considered).

In this paper, we present the core as the expected value computed over all possible string continuations, each weighted by its LLM-induced probability. This choice links homogenization to a quantity that current evaluations already measure: variance. Using the mode instead of the mean would yield a different deviance (mode deviation).

Inspired by (Leinster, 2024) (which shows that diversity admits a parametrized family of measures), we develop several profiles of normativity in Appendix A as a parametrized family (that connect many statistics, such as the mean and mode). Future work will explore other characterizations of normativity in this family.

3.3 LLMs as trees of strings

Let $\{t_a, t_b, \ldots\}$ denote the finite token alphabet, with special tokens $\perp$ (start-of-sequence) and $\top$ (end-of-sequence). A **string** is a finite sequence of tokens beginning with $\perp$; a **trajectory** is a string

[4] To paraphrase (Cheng, 2022), abstraction is about making precise the different senses in which different things can be valid.

ending with $\top$. We write prompts, continuations, and trajectories as:

$$x_p = \bot t_1 \dots t_p$$
$$x_{p+k} = x_p t_{p+1} \dots t_{p+k}$$
$$y = x_T = x_{T-1} \top$$

We denote the set of strings that are continuations of a prompt string x_p as $\mathsf{Str}(x_p)$. The unprompted scenario corresponds to $x_p = \bot$. Then, we write the set of all strings as $\mathsf{Str} := \mathsf{Str}(\bot)$. Similarly, we denote the set of strings that are trajectories of a prompt string x_p as $\mathsf{Str}_\top(x_p) \subseteq \mathsf{Str}(x_p)$, and the set of all trajectories as $\mathsf{Str}_\top := \mathsf{Str}_\top(\bot) \subseteq \mathsf{Str}$.

Any LLM induces a tree on Str: the root is $\bot$, each node is a string, the leaves are trajectories, and the edges connect strings by their next-token continuations with probability $p(t_{p+1}|x_p)$. Probabilities chain and decompose as $p(y|x_p) = p(x_{p+k}|x_p)p(y|x_{p+k})$.

For any prompt x, we have a probability mass function on the trajectories for any particular prompt (Bradley and Vigneaux, 2025). For simplicity, we assume all terminal strings finish within a finite context window [5]. We can then write:

$$\sum_{y \in \mathsf{Str}_\top(x_p)} p(y|x_p) = 1 \tag{1}$$

3.4 Structure-awareness

For a string $x \in \mathsf{Str}$, the degree of **structure compliance** is:

$$\alpha_i : \mathsf{Str} \to [0, 1] \tag{2}$$

We can think that, for a given x string, ideal compliance corresponds to $\alpha_i(x) = 1$, and no compliance corresponds to $\alpha_i(x) = 0$.

We can consider many structures simultaneously. We call a **system** the collection of structures of interest. We define the **system compliance** as a vector of compliances across particular structures:

$$\Lambda_n(x) := (\alpha_1(x), \dots, \alpha_n(x)) \tag{3}$$

[5] This is a simplifying assumption for exposition. To be fully precise, we would instead formulate this as $y \in Terminating(x_p)$ where $\mathsf{Str}_\top(x_p) \subseteq Terminating(x_p)$. We will provide a deeper analysis of the ambiguity of terminating unfinished strings in future work. Refer to (Bradley and Vigneaux, 2025) for the theoretical foundation for LLMs as trees of strings.

To enable easy comparisons, we define operators[6] that aggregate compliance into scalar **system scores** and **difference scores**:

$$\|\Lambda_n(x)\|_\Lambda,\ \|\Lambda_n(x_r) - \Lambda_n(x_q)\|_\theta \ \in\ [0, 1] \tag{4}$$

3.5 Incorporating string statistics

For a given structure and an LLM, we can reason about its expected structural compliance. We call this the **structure core**:

$$\langle \alpha_i \rangle = \sum_{y \in \mathsf{Str}_\top} p(y)\alpha_i(y) \tag{5}$$

Similarly, we can reason about the expected system compliance as the **system core**:

$$\langle \Lambda_n \rangle = \sum_{y \in \mathsf{Str}_\top} p(y)\Lambda_n(y) \tag{6}$$

Leveraging these definitions, we can reason about the deviation from the expected system compliance. This would constitute a set of deviations, one deviation for each structure. The **orientation** (Ahmed, 2006) of a given string relative to the given system core is:

$$\theta_n(x) = \Lambda_n(x) - \langle \Lambda_n \rangle \tag{7}$$

To summarize non-normativity as a single number, we leverage Equation 4 to define the **deviance**:

$$\|\theta_n(x)\|_\theta = \partial_n(x) \in [0, 1] \tag{8}$$

3.6 What about prompting?

We can generalize our framework to account for all prompts by making explicit the **conditioning on a given prompt** x_p:

$$\begin{aligned} \langle \alpha_i \rangle(x_p) &= \sum_{y \in \mathsf{Str}_\top(x_p)} p(y|x_p)\alpha_i(y) \\ \langle \Lambda_n \rangle(x_p) &= \sum_{y \in \mathsf{Str}_\top(x_p)} p(y|x_p)\Lambda_n(y) \\ \theta_n(x|x_p) &= \Lambda_n(x) - \langle \Lambda_n \rangle(x_p) \end{aligned} \tag{9}$$

The conditional probabilities under different prompts may differ substantially. Different prompts collapse to different modes (Zhang et al., 2025a). **We can think that a given prompt induces its own normativity**.

In D, we discuss how prompting can be interpreted as dynamics. Figure 1 visualizes how the conditioned system cores $\langle \Lambda_n \rangle(x_p)$ establish the frames of reference for diversity and deviance.

[6] While system compliance is formulated as a vector, this generalizes to other structures with appropriate operators. See A.

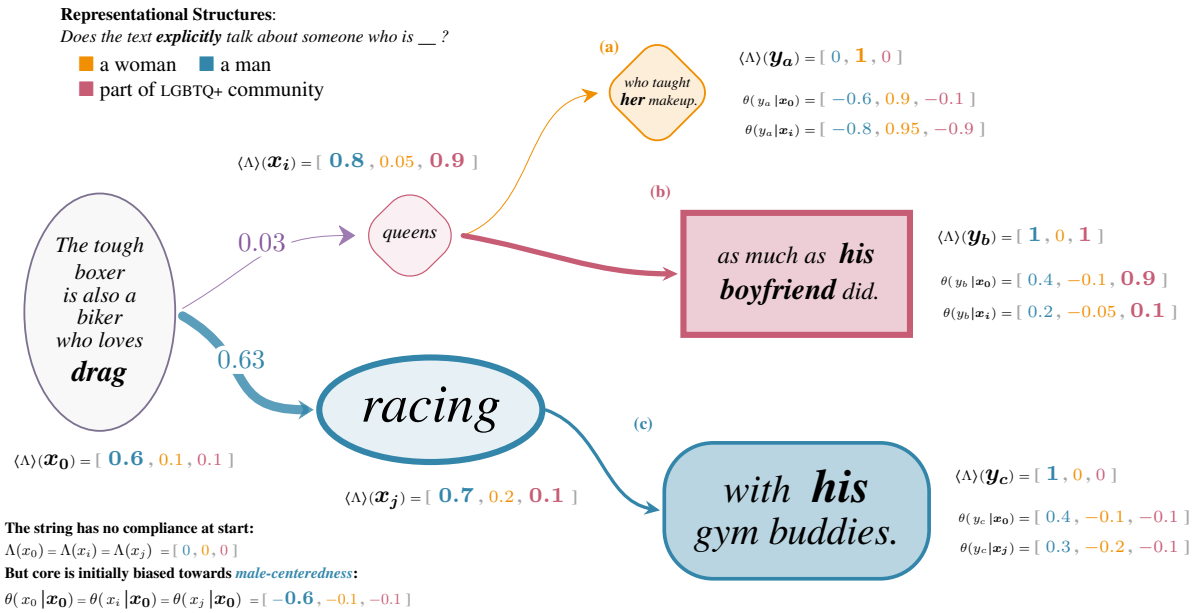

Figure 1: **Illustration of how system cores and orientations *evolve* through trajectories.** In the example above, our system has *sub-community representation* structures. We can calculate each compliance by asking a *judge LLM* (Zhu et al., 2025) to rate from [0,1] based on whether the *community* subgroup is *explicitly* represented in the string of text. Though initially *ambiguous*, the phrasing of the prompt may invite *stereotyping*, biasing continuations to turn *male-centered*. Trajectory (c) constitutes the *greediest* trajectory out of the three. The *unmarked* prompt (Gillespie, 2024) defaults to the normative path. Although trajectory (b) is considered fairly deviant *relative to* x_0 (unmarked prompt), it is much less so *relative to* x_i (...drag queens), exemplifying how diversity itself is fundamentally relative. The *branching point* at "drag" *rarely* extends text into the "queens" *subtree*. Still, whenever it does, the resulting trajectories are biased to mention members of the LGBTQ+ community. Notice how even though the "queens" subtree is deviant relative to the prompt in ■ dimension, it is still normative in ■ dimension and remains biased against ■-*compliant* trajectories like (a).

4 Homogenization

When alignment optimizes for the core, outputs cluster, variance shrinks, and the long tails disappear. We call this homogenization: the collapse of diversity toward dominant modes.

Formally, we can consider the expected deviance and the deviance variance:

$$\mathbb{E}_{y\sim p(\cdot|x_p)}[\partial_n] = \sum_{y\in \mathsf{Str}_\top(x_p)} p(y|x_p)\ \partial_n(y|x_p)$$
$$\mathrm{Var}_{y\sim p(\cdot|x_p)}[\partial_n] = (\mathbb{E}[\partial_n^2] - \mathbb{E}[\partial_n]^2)_{y\sim p(\cdot|x_p)} \tag{10}$$

Then, we can see homogenization as **minimizing** all deviance [7]:

$$\mathbb{E}_{y\sim p(\cdot|x_p)}[\partial_n] \ \mapsto\ 0 \qquad \mathrm{Var}_{y\sim p(\cdot|x_p)}[\partial_n] \ \mapsto\ 0 \tag{11}$$

Given a system core $\langle\Lambda_n\rangle$, we can normalize its structures as $\langle\overline{\alpha}_{\mathrm{norm}_i}\rangle := \frac{\langle\alpha_i\rangle(x_p)}{\sum_j^n \langle\alpha_j\rangle(x_p)}$. Then, we can compute the core entropy:

$$H(\langle\Lambda_n\rangle) = -\sum_{i=1}^{n} \langle\overline{\alpha}_{\mathrm{norm}_i}\rangle \log(\langle\overline{\alpha}_{\mathrm{norm}_i}\rangle) \tag{12}$$

[7] Here, $\mapsto$ represents "is pushed towards"

Then, we can also think of homogenization as making the system core **more uneven**. When the core has low entropy, fewer structures dominate:

$$H(\langle\Lambda_n\rangle) \ \mapsto\ 0 \tag{13}$$

5 Xeno-reproduction

Where homogenization collapses variation into dominant modes, xeno-reproduction seeks the long tails. Where traces vanish, it amplifies them. Concretely, xeno-reproduction is **structure-aware diversity pursuit**: a non-objective search (Lehman and Stanley, 2011) that can be augmented with fairness biases and explicit constraints.

5.1 Distribution-level formulation

We score interventions through the intervention variable w that encompasses any[8] mechanism affecting the effective distribution of trajectories.

[8] We consider anything that depends on $p(y|x_p, w)$ to be parameterized by w as well. For instance, $\langle\Lambda_n\rangle$ would be parametrized as $\langle\Lambda_n\rangle(x_p, w)$, but for readability we just write $\langle\Lambda_n\rangle(w)$, folding the prompting into the interventional variable.

We write w_0 for the unintervened conditions (the baseline).

5.1.1 Scoring diversity

We would like to evaluate how much more diversity-seeking our choice of w is compared to the baseline.

On the one hand, we can think of promoting diversity as inducing a new core that is different from the old one:

$$\mathsf{score}_{\mathsf{explore}}(w) = \|\langle\Lambda_n\rangle(w) - \langle\Lambda_n\rangle(w_0)\|_\theta \quad (14)$$

On the other hand, the new core should not be excessively dominant. We can think of promoting diversity as guiding output strings to diverge from any system core, and also be deviant in their own way:

$$\mathsf{score}_{\mathsf{diverge}}(w) = \lambda_{\mathbb{E}}\mathbb{E}[\partial_n](w) + \lambda_{\mathrm{Var}}\,\mathrm{Var}[\partial_n](w) \quad (15)$$

Our **diversity score** ρ_d would then be a λ-weighted sum:

$$\rho_d(w) = \lambda_{d_0}\mathsf{score}_{\mathsf{explore}}(w) + \lambda_{d_1}\mathsf{score}_{\mathsf{diverge}}(w) \quad (16)$$

5.1.2 Scoring fairness

We frame diversity as fairness (Verma and Rubin, 2018). Structural parity means no single structure dominates. Justice-oriented fairness (Rawls, 1971; Mittelstadt et al., 2023) means interventions maximally benefit the least advantaged. Core evenness operationalizes this: when all structures have equal expected compliance, no pattern is privileged.

We evaluate how even the system core is:

$$\mathsf{score}_{\mathsf{even}}(w) = H(\langle\Lambda_n\rangle(w)) \quad (17)$$

We would also like to evaluate how much our choice of w inverts the normative ordering of the structure cores induced by w_0. To do so, we can leverage the relative-order sign:

$$s_{i,j}(w) = \mathrm{sign}(\langle\alpha_i\rangle(w) - \langle\alpha_j\rangle(w)) \;\in\; \{-,0,+\} \quad (18)$$

We can score the invertedness of the normative order (Equation D.1) as:

$$\mathsf{score}_{\mathsf{inverted}}(w) = \binom{n}{2}^{-1}\sum_{1\le i<j\le n}\mathbf{1}[s_{i,j}(w) \neq s_{i,j}(w_0)] \quad (19)$$

Our **fairness score** ρ_f would then be a λ-weighted sum:

$$\rho_f(w) = \lambda_{f_0}\mathsf{score}_{\mathsf{even}}(w) + \lambda_{f_1}\mathsf{score}_{\mathsf{inverted}}(w) \quad (20)$$

5.1.3 Scoring adherence to constraints

To be explicit and intentional, we need to consider constraints (Eguchi, 2024). We can define systems that prescribe the structures that we would like to target, conserve and avoid. We would like to score how much our choice of w affects the adherence to those constraints. Our **constraint score** ρ_c would be a λ-weighted sum:

$$\begin{aligned}\rho_c(w) = \lambda_{c_0}\|\langle\Lambda_{\mathsf{target}}\rangle(w)\|_\Lambda - \lambda_{c_1}\|\langle\Lambda_{\mathsf{avoid}}\rangle(w)\|_\Lambda \\ -\lambda_{c_2}\|\langle\Lambda_{\mathsf{conserve}}\rangle(w) - \langle\Lambda_{\mathsf{conserve}}\rangle(w_0)\|_\theta\end{aligned} \quad (21)$$

5.1.4 Xeno-reproduction as search over interventions

The **intervention score** ρ_χ is a λ-weighted sum:

$$\rho_\chi(w) = \lambda_d\rho_d(w) + \lambda_f\rho_f(w) + \lambda_c\rho_c(w) \quad (22)$$

We formulate xeno-reproduction as the **search over interventions**:

$$w \sim \pi(w) \;\propto\; e^{\beta_\rho\rho_\chi(w)} \quad (23)$$

where β_ρ is a tunable temperature parameter.

By sampling the intervention variable and applying it, we generate trajectories:

$$\mathbb{E}_{w\sim\pi(w)}[p(y|w)] = \int \pi(w)\,p(y|w)\,dw \quad (24)$$

We also develop a complementary **trajectory-level formulation** that assigns rewards to individual outputs, enabling more tractable implementations through sample-based approximations. This formulation is detailed in E.

6 Related Work

Our framework enters ongoing conversations about how to make AI systems deviate productively.

Active Divergence (Berns et al., 2023; Broad et al., 2021; Berns, 2025; Berns and Colton, 2020; Tahiroglu and Wyse, 2024; Esling et al., 2022; Cole et al., 2025) also aims to disorient (Ahmed, 2006). However, Active Divergence maximizes raw novelty in artistic contexts, whereas xeno-reproduction addresses homogenization through structures and is oriented towards AI safety rather than computational creativity.

Interpretability will help us understand how structures relate to models' internals. At a foundational layer, both fields converge on representation bias: the phenomenon where some signals are

represented more strongly than others, even when equally relevant (Lampinen et al., 2024, 2025).

Reinforcement Learning and xeno-reproduction both leverage exploration (Song et al., 2025; Yao et al., 2023). Search algorithms like AlphaSAGE (Chen et al., 2025) and Quality-Diversity (Pugh et al., 2016) are promising directions.

C situates our framework within common linguistic diversity metrics. Our framework can accommodate existing metrics using a common language.

7 Alternative views

Skepticism of technical solutions. Some authors argue (Wachter et al., 2021; Davis and Williams, 2025; Green and Viljoen, 2020) that technical interventions are inappropriate for what is fundamentally a social justice problem. Better interventions might focus on institutional change, community participation, or stopping AI development altogether (Goldfarb, 2024). This critique is serious. Xeno-reproduction could fall into the solutionism trap (Selbst et al., 2019). We proceed anyway, not because technical solutions are sufficient, but because they are one lever among many.

Diversity can be risky. Open-ended search brings unpredictability, uncontrollability, and potential misalignment (Sheth et al., 2025). The effect on LLM performance remains an open question; our framework can encode performance metrics as target structures, but trade-offs require empirical investigation. Despite these risks, open-endedness could ultimately make AI safety more robust (Hughes et al., 2024; Taleb, 2013).

8 Conclusion

This paper presents a case for diversity and identifies xeno-reproduction as a strategy that intentionally promotes it. This paper also presents an expressive framework for accounting for the structures of strings and their corresponding statistics. This is just an initial step towards scholarships that seriously theorize diversity and foreground its impact on people at the margins.

Limitations

Diversity is complex. Our framework is not complete; it is a starting point. Significant collaboration will be required to address homogenization effectively.

Specification of structures. The choice of structures is always opinionated. We need a taxonomy of structure types, specifying how compliance could be estimated. Future work will align our framework with emerging research in computational learning theory and language generation that formalizes hallucination trade-offs[9] (Kalavasis et al., 2025b).

Computational tractability. Calculating the system core exactly requires summing over $y \in \mathsf{Str}_{\top}(x_p)$, which is intractable. Tractable approximation methods are needed, possibly using smart sampling (Macar et al., 2025) or carefully designed prompting (Zhang et al., 2025a).

Connecting to evaluations. Our framework provides a language for evaluation: reporting what diversity is lost or preserved through alignment. We have intuition for how to operationalize this (LLM-as-judge (Zhu et al., 2025), structure classifiers), but connecting to existing diversity benchmarks (Jiang et al., 2025; Zhang et al., 2025b) remains future work.

Ethical tensions. Who should define structures? Community participation is needed. Is visibility always beneficial? Minoritized populations sometimes prefer opacity as protection. Consent-based approaches are needed.

Call to action

We call for AI Safety to:

- Integrate homogenization into threat models and evaluations.
- Be explicit about context. Diversity in what? For whom?
- Engage seriously with critical theory: Queer theory, Black studies, Postcolonial studies.

Impact Statement

This paper introduces a formal framework to center diversity in AI Safety. However, there are important risks. **The same methods that aim to amplify diversity could be used to squash, exploit, and control it.** Additionally, any formalization of diversity also risks reproducing the exclusions we aim to address.

References

Dhruv Agarwal, Mor Naaman, and Aditya Vashistha. 2025. Ai suggestions homogenize writing toward

[9] See B for discussion.

western styles and diminish cultural nuances. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, CHI '25, page 1–21. ACM.

Sara Ahmed. 2006. *Queer Phenomenology: Orientations, Objects, Others*. Duke University Press.

Sumukh K Aithal, Pratyush Maini, Zachary C Lipton, and J Zico Kolter. 2024. Understanding hallucinations in diffusion models through mode interpolation. *URL https://arxiv. org/abs/2406.09358*, 2406.

Hunt Allcott, Luca Braghieri, Sarah Eichmeyer, and Matthew Gentzkow. 2020. The welfare effects of social media. 110(3):629–676.

Isobel Barry and Elise Stephenson. 2025. The gendered, epistemic injustices of generative ai. *Australian Feminist Studies*, 40(123):1–21.

Paul Beamish and Vanessa Hasse. 2022. The importance of rare events and other outliers in global strategy research. *Global Strategy Journal*, 12:697–713.

Yoshua Bengio, Sören Mindermann, Daniel Privitera, and 1 others. 2025. International ai safety report. Technical Report DSIT 2025/001, UK Department for Science, Innovation and Technology. First International AI Safety Report, published January 2025.

Franco Berardi. 2017. *Futurability: The Age of Impotence and the Horizon of Possibility*. Verso.

J.-F. Bercher. 2011. Escort entropies and divergences and related canonical distribution. *Physics Letters A*, 375(33):2969–2973.

S Berns. 2025. *Diversity in Generative Machine Learning to Enhance Creative Applications*. Ph.D. thesis, Queen Mary University of London.

Sebastian Berns and Simon Colton. 2020. Bridging generative deep learning and computational creativity. In *Proceedings of the 11th International Conference on Computational Creativity (ICCC'20)*, pages 406–409.

Sebastian Berns, Simon Colton, and Christian Guckelsberger. 2023. Towards mode balancing of generative models via diversity weights. *Preprint*, arXiv:2304.11961.

T.M. Bettcher. 2025. *Beyond Personhood: An Essay in Trans Philosophy*. University of Minnesota Press.

Alexander Betts, Louise Bloom, Josiah Kaplan, and Naohiko Omata. 2017. *Refugee Economies: Forced Displacement and Development*. Oxford University Press, Oxford.

Dila Ram Bhandari, Kapil Shah, and Aayan Bhandari. 2024. The power of outliers in research: What actually works, and does it matter? *Pravaha*, 30(1):84–91.

Rishi Bommasani, Kathleen A. Creel, Ananya Kumar, Dan Jurafsky, and Percy S Liang. 2022. Picking on the same person: Does algorithmic monoculture lead to outcome homogenization? In *Advances in Neural Information Processing Systems*, volume 35, pages 3663–3678. Curran Associates, Inc.

Geoffrey C. Bowker and Susan Leigh Star. 1999. *Sorting Things Out: Classification and Its Consequences*. Inside Technology. MIT Press, Cambridge, MA; London, England. First edition. Also available as MIT Press paperback, 2000, ISBN 978-0-262-52295-3; eISBN 978-0-262-26907-0.

Tai-Danae Bradley and Juan Pablo Vigneaux. 2025. The magnitude of categories of texts enriched by language models. *Preprint*, arXiv:2501.06662.

Terence Broad, Sebastian Berns, Simon Colton, and Mick Grierson. 2021. Active divergence with generative deep learning–a survey and taxonomy. *arXiv preprint arXiv:2107.05599*.

Benjamin S. Bucknall. 2022. Current and near-term ai as a potential existential risk factor. *Preprint*, arxiv:2209.10604.

Binqi Chen, Hongjun Ding, Ning Shen, Jinsheng Huang, Taian Guo, Luchen Liu, and Ming Zhang. 2025. Alphasage: Structure-aware alpha mining via gflownets for robust exploration. *Preprint*, arXiv:2509.25055.

Eugenia Cheng. 2022. *The Joy of Abstraction: An Exploration of Math, Category Theory, and Life*. Cambridge University Press.

Matthew Cobbinah, Henry Nunoo-Mensah, Prince Ebenezer Adjei, Francisca Adoma Acheampong, Isaac Acquah, Eric Tutu Tchao, Andrew Selasi Agbemenu, Jerry John Kponyo, and Emmanuel Abaidoo. 2025. Diversity in stable gans: A systematic review of mode collapse mitigation strategies. *Engineering Reports*, 7(6):e70209.

Mark Coeckelbergh. 2023. Narrative responsibility and artificial intelligence: How ai challenges human responsibility and sense-making. *AI & SOCIETY*, 38(6):2437–2450.

Adam Cole, Gregor Petrikovič, and Mick Grierson. 2025. Me vs. you: Wrestling with ai's limits through queer experimental filmmaking. In *Proceedings of the 2025 Conference on Creativity and Cognition*, pages 836–841.

Beth Coleman. 2021. Technology of the surround. *Catalyst: Feminism, Theory, Technoscience*, 7(2):1–21.

Chelsea N Cook, Angela R Freeman, James C Liao, and Lisa A Mangiamele. 2021. The philosophy of outliers: reintegrating rare events into biological science. *Integrative and Comparative Biology*, 61(6):2191–2198.

Manuel Cossio. 2025. A comprehensive taxonomy of hallucinations in large language models. *Preprint*, arXiv:2508.01781.

Jenny L Davis and Apryl Williams. 2025. Repair and redress: A research program for algorithmic futures.

Jeffrey Dean and Luiz André Barroso. 2013. The tail at scale. *Communications of the ACM*, 56(2):74–80.

Fermin Moscoso del Prado Martin. 2024. Measuring grammatical diversity from small corpora: Derivational entropy rates, mean length of utterances, and annotation invariance. *Preprint*, arXiv:2412.06095.

Benjamin Edwards, Steven Hofmeyr, and Stephanie Forrest. 2016. Hype and heavy tails: A closer look at data breaches. *Journal of Cybersecurity*, 2(1):3–14.

Shinto Eguchi. 2024. Information geometry for maximum diversity distributions. *Preprint*, arXiv:2412.03835.

Philippe Esling and 1 others. 2022. Challenges in creative generative models for music: a divergence maximization perspective. *arXiv preprint arXiv:2211.08856*.

Louis Estève, Marie-Catherine de Marneffe, Nurit Melnik, Agata Savary, and Olha Kanishcheva. 2025. A survey of diversity quantification in natural language processing: The why, what, where and how. *Preprint*, arXiv:2507.20858.

Facebook. 2021. Facebook response: Sri lanka human rights impact assessment.

Paul Farmer. 2004. An anthropology of structural violence. *Current Anthropology*, 45(3):305–325.

Sina Fazelpour and Meica Magnani. 2025. Aspirational affordances of ai. *Preprint*, arXiv:2504.15469.

Shangbin Feng, Wenhao Yu, Yike Wang, Hongming Zhang, Yulia Tsvetkov, and Dong Yu. 2025. Don't throw away your pretrained model. *Preprint*, arXiv:2510.09913.

Matthew Finlayson, John Hewitt, Alexander Koller, Swabha Swayamdipta, and Ashish Sabharwal. 2023. Closing the curious case of neural text degeneration. *Preprint*, arXiv:2310.01693.

Marie J. C. Forgeard. 2013. Perceiving benefits after adversity: The relationship between self-reported posttraumatic growth and creativity. *Psychology of Aesthetics, Creativity, and the Arts*, 7(3):245–264.

Dan Friedman and Adji Bousso Dieng. 2023. The vendi score: A diversity evaluation metric for machine learning. *Preprint*, arXiv:2210.02410.

Johan Galtung. 1969. Violence, peace, and peace research. *Journal of Peace Research*, 6(3):167–191.

Tarleton Gillespie. 2024. Generative ai and the politics of visibility. *Big Data & Society*, 11(2):20539517241252131.

Trystan S. Goetze. 2018. Hermeneutical dissent and the species of hermeneutical injustice. *Hypatia*, 33(1):73–90.

Avi Goldfarb. 2024. Pause artificial intelligence research? understanding ai policy challenges. *Canadian Journal of Economics/Revue canadienne d'économique*, 57(2):363–377.

Gayatri Gopinath. 2005. *Impossible Desires: Queer Diasporas and South Asian Public Cultures*. Duke University Press, Durham, NC.

Deborah B. Gould. 2009. *Moving Politics: Emotion and ACT UP's Fight against AIDS*. University of Chicago Press, Chicago.

Ben Green and Salomé Viljoen. 2020. Algorithmic realism: expanding the boundaries of algorithmic thought. In *Proceedings of the 2020 conference on fairness, accountability, and transparency*, pages 19–31.

Jingyi Gu, Xuan Zhang, and Guiling Wang. 2025. Beyond the norm: A survey of synthetic data generation for rare events. *Preprint*, arXiv:2506.06380.

Yanzhu Guo, Guokan Shang, and Chloé Clavel. 2025. Benchmarking linguistic diversity of large language models. *Preprint*, arXiv:2412.10271.

Yanzhu Guo, Guokan Shang, Michalis Vazirgiannis, and Chloé Clavel. 2024a. The curious decline of linguistic diversity: Training language models on synthetic text. *Preprint*, arXiv:2311.09807.

Yufei Guo, Muzhe Guo, Juntao Su, Zhou Yang, Mengqiu Zhu, Hongfei Li, Mengyang Qiu, and Shuo Shuo Liu. 2024b. Bias in large language models: Origin, evaluation, and mitigation. *Preprint*, arXiv:2411.10915.

Jeremy Hadfield. 2023. Why ai ethics needs conceptual engineers. Imaginaries (Substack).

Aimi Hamraie and Kelly Fritsch. 2019. Crip technoscience manifesto. *Catalyst: Feminism, Theory, Technoscience*, 5(1):1–33.

Byung-Chul Han. 2024. *The Crisis of Narration*. Polity Press.

Jacqueline Harding and Cameron Domenico Kirk-Giannini. 2025. What is ai safety? what do we want it to be? *Preprint*, arXiv:2505.02313.

Stefano Harney and Fred Moten. 2013. *The Undercommons: Fugitive Planning & Black Study*. Minor Compositions.

H. Hester. 2018. *Xenofeminism*. Theory Redux. Polity Press.

Bas Hofstra, Vivek V Kulkarni, Sebastian Munoz-Najar Galvez, Bryan He, Dan Jurafsky, and Daniel A McFarland. 2020. The diversity–innovation paradox in science. *Proceedings of the National Academy of Sciences*, 117(17):9284–9291.

bell hooks. 1990. Choosing the margin as a space of radical openness. In *Yearning: Race, Gender, and Cultural Politics*, pages 145–153. South End Press, Boston.

Lei Huang, Weijiang Yu, Weitao Ma, Weihong Zhong, Zhangyin Feng, Haotian Wang, Qianglong Chen, Weihua Peng, Xiaocheng Feng, Bing Qin, and Ting Liu. 2025. A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions. *ACM Transactions on Information Systems*, 43(2):1–55.

Linus Ta-Lun Huang and Tsung-Ren Huang. 2025. Generative bias: widespread, unexpected, and uninterpretable biases in generative models and their implications. *AI & SOCIETY*, pages 1–13.

Yiwen Huang, Aaron Gokaslan, Volodymyr Kuleshov, and James Tompkin. 2024. The gan is dead; long live the gan! a modern gan baseline. In *Advances in Neural Information Processing Systems*, volume 37, pages 44177–44215. Curran Associates, Inc.

Edward Hughes, Michael Dennis, Jack Parker-Holder, Feryal Behbahani, Aditi Mavalankar, Yuge Shi, Tom Schaul, and Tim Rocktaschel. 2024. Open-endedness is essential for artificial superhuman intelligence. *Preprint*, arXiv:2406.04268.

Atif Hussain. 2024. Voice and ai: The subaltern's challenge. Medium.

Shomik Jain, Jack Lanchantin, Maximilian Nickel, Karen Ullrich, Ashia Wilson, and Jamelle Watson-Daniels. 2025. Llm output homogenization is task dependent. *Preprint*, arXiv:2509.21267.

Sheila Jasanoff. 2007. Technologies of humility. *Nature*, 450(7166):33–33.

Liwei Jiang, Yuanjun Chai, Margaret Li, Mickel Liu, Raymond Fok, Nouha Dziri, Yulia Tsvetkov, Maarten Sap, Alon Albalak, and Yejin Choi. 2025. Artificial hivemind: The open-ended homogeneity of language models (and beyond). *Preprint*, arXiv:2510.22954.

Feng Ju, Zeyu Qin, Rui Min, Zhitao He, Lingpeng Kong, and Yi R. Fung. 2025. Reasoning path divergence: A new metric and curation strategy to unlock llm diverse thinking. *Preprint*, arXiv:2510.26122.

Daniel Jurafsky and James H. Martin. 2009. *Speech and Language Processing*, 2nd edition. Pearson Prentice Hall.

Adam Tauman Kalai, Ofir Nachum, Santosh S. Vempala, and Edwin Zhang. 2025. Why language models hallucinate. *Preprint*, arXiv:2509.04664.

Alkis Kalavasis, Anay Mehrotra, and Grigoris Velegkas. 2025a. On characterizations for language generation: Interplay of hallucinations, breadth, and stability. *Preprint*, arXiv:2412.18530.

Alkis Kalavasis, Anay Mehrotra, and Grigoris Velegkas. 2025b. On the limits of language generation: Trade-offs between hallucination and mode collapse. *Preprint*, arXiv:2411.09642.

Atoosa Kasirzadeh. 2025. Two types of ai existential risk: Decisive and accumulative.

Jared Katzman, Angelina Wang, Morgan Scheuerman, Su Lin Blodgett, Kristen Laird, Hanna Wallach, and Solon Barocas. 2023. Taxonomizing and measuring representational harms: A look at image tagging. *Preprint*, arXiv:2305.01776.

Kelly Kendro, Jeffrey Maloney, and Scott Jarvis. 2025. Do llms produce texts with "human-like" lexical diversity? *Unpublished manuscript / ResearchGate*.

Jon Kleinberg and Sendhil Mullainathan. 2024. Language generation in the limit. In *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*.

Noam Kolt. 2024. Algorithmic black swans. 101:1177–1240.

Tomek Korbak, Mikita Balesni, Elizabeth Barnes, Yoshua Bengio, Joe Benton, Joseph Bloom, Mark Chen, Alan Cooney, Allan Dafoe, Anca Dragan, Scott Emmons, Owain Evans, David Farhi, Ryan Greenblatt, Dan Hendrycks, Marius Hobbhahn, Evan Hubinger, Geoffrey Irving, Erik Jenner, and 22 others. 2025. Chain of thought monitorability: A new and fragile opportunity for ai safety. *Preprint*, arXiv:2507.11473.

Andrew Kyle Lampinen, Stephanie C. Y. Chan, and Katherine Hermann. 2024. Learned feature representations are biased by complexity, learning order, position, and more. *Preprint*, arXiv:2405.05847.

Andrew Kyle Lampinen, Stephanie CY Chan, Yuxuan Li, and Katherine Hermann. 2025. Representation biases: will we achieve complete understanding by analyzing representations? *arXiv preprint arXiv:2507.22216*.

Kang-il Lee, Hyukhun Koh, Dongryeol Lee, Seunghyun Yoon, Minsung Kim, and Kyomin Jung. 2025. Generating diverse hypotheses for inductive reasoning. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 8461–8474, Albuquerque, New Mexico. Association for Computational Linguistics.

Messi H. J. Lee. 2025. Examining the robustness of homogeneity bias to hyperparameter adjustments in gpt-4. *Preprint*, arXiv:2501.02211.

Joel Lehman and Kenneth O Stanley. 2011. Novelty search and the problem with objectives. In *Genetic programming theory and practice IX*, pages 37–56. Springer.

Tom Leinster. 2024. Entropy and diversity: The axiomatic approach. *Preprint*, arXiv:2012.02113.

Rafael P Leitão, Jansen Zuanon, Sébastien Villéger, Stephen E Williams, Christopher Baraloto, Claire Fortunel, Fernando P Mendonça, and David Mouillot. 2016. Rare species contribute disproportionately to the functional structure of species assemblages. *Proceedings of the Royal Society B: Biological Sciences*, 283(1828):20160084.

Chaozhuo Li, Pengbo Wang, Chenxu Wang, Litian Zhang, Zheng Liu, Qiwei Ye, Yuanbo Xu, Feiran Huang, Xi Zhang, and Philip S. Yu. 2025. Loki's dance of illusions: A comprehensive survey of hallucination in large language models. *Preprint*, arXiv:2507.02870.

Cheuk Ting Li and Farzan Farnia. 2023. Mode-seeking divergences: theory and applications to gans. In *International Conference on Artificial Intelligence and Statistics*, pages 8321–8350. PMLR.

Jiwei Li, Michel Galley, Chris Brockett, Jianfeng Gao, and Bill Dolan. 2016. A diversity-promoting objective function for neural conversation models. In *Proceedings of the 2016 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 110–119. Association for Computational Linguistics.

Paola Lopez. 2021. Bias does not equal bias: A socio-technical typology of bias in data-based algorithmic systems. *Internet Policy Review*, 10(4):1–29.

Uzay Macar, Paul C. Bogdan, Senthooran Rajamanoharan, and Neel Nanda. 2025. Thought branches: Interpreting llm reasoning requires resampling. *Preprint*, arXiv:2510.27484.

D. McQuillan. 2022. *Resisting AI: An Anti-fascist Approach to Artificial Intelligence*. Bristol University Press.

Tao Meng, Ninareh Mehrabi, Palash Goyal, Anil Ramakrishna, Aram Galstyan, Richard Zemel, Kai-Wei Chang, Rahul Gupta, and Charith Peris. 2024. Attribute controlled fine-tuning for large language models: A case study on detoxification. *Preprint*, arXiv:2410.05559.

Mikhail Mironov and Liudmila Prokhorenkova. 2025. Measuring diversity: Axioms and challenges. *Preprint*, arXiv:2410.14556.

Brent Mittelstadt, Sandra Wachter, and Chris Russell. 2023. The unfairness of fair machine learning: Levelling down and strict egalitarianism by default. *Michigan Technology Law Review*, 30(1):1–76.

Anamika Modok. 2023. Role of social media in inciting the genocidal acts: A case study on myanmar's rohingya. *Contemporary Challenges: The Global Crime, Justice and Security Journal*, 4.

Shakir Mohamed, Marie-Therese Png, and William Isaac. 2020. Decolonial ai: Decolonial theory as sociotechnical foresight in artificial intelligence. *Philosophy & Technology*, 33(4):659–684.

Kibum Moon, Adam E Green, and Kostadin Kushlev. 2025. Homogenizing effect of large language models (llms) on creative diversity: An empirical comparison of human and chatgpt writing. *Computers in Human Behavior: Artificial Humans*, page 100207.

Robert Morain and Dan Ventura. 2025. Is prompt engineering the creativity knob for large language models? In *Proceedings of the 16th International Conference for Computational Creativity*.

Evgeny Morozov. 2024. The ai we deserve.

Sonia Krishna Murthy, Tomer Ullman, and Jennifer Hu. 2025. One fish, two fish, but not the whole sea: Alignment reduces language models' conceptual diversity. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, page 11241–11258. Association for Computational Linguistics.

Max Peeperkorn, Tom Kouwenhoven, Dan Brown, and Anna Jordanous. 2024. Is temperature the creativity parameter of large language models? *Preprint*, arXiv:2405.00492.

Max Peeperkorn, Tom Kouwenhoven, Dan Brown, and Anna Jordanous. 2025. Mind the gap: Conformative decoding to improve output diversity of instruction-tuned large language models. *arXiv preprint arXiv:2507.20956*.

Andrew J. Peterson. 2025. Ai and the problem of knowledge collapse. *AI & SOCIETY*, 40(5):3249–3269.

Krishna Pillutla, Lang Liu, John Thickstun, Sean Welleck, Swabha Swayamdipta, Rowan Zellers, Sewoong Oh, Yejin Choi, and Zaid Harchaoui. 2023. Mauve scores for generative models: Theory and practice. *Preprint*, arXiv:2212.14578.

P Preciado. 2013. *Testo Junkie: Sex, Drugs, and Biopolitics in the Pharmacopornographic Era*. G - Reference,Information and Interdisciplinary Subjects Series. Feminist Press at the City University of New York.

Adrian Preda. 2025. Special report: Ai-induced psychosis: a new frontier in mental health.

Justin K Pugh, Lisa B Soros, and Kenneth O Stanley. 2016. Quality diversity: A new frontier for evolutionary computation. *Frontiers in Robotics and AI*, 3:40.

Rizky Putra, Ayu Kartika, and Budi Santoso. 2024. Solving long-tail detection for autonomous vehicles. *Authorea Preprints*.

Salsa Della Guitara Putri, Eko Priyo Purnomo, and Tiara Khairunissa. 2024. Echo chambers and algorithmic bias: The homogenization of online culture in a smart society. In *SHS Web of Conferences*, volume 202, page 05001. EDP Sciences.

John Rawls. 1971. *A Theory of Justice*. Harvard University Press, Cambridge, MA.

Karoline Reinhardt. 2020. Between identity and ambiguity: some conceptual considerations on diversity. *Symposion*, 7(2):261–283.

Ermelinda Rodilosso. 2024. Filter bubbles and the unfeeling: How ai for social media can foster extremism and polarization. *Philosophy & Technology*, 37(2):71.

Ihor Rudko and Aysan Bashirpour Bonab. 2025. Chatgpt is incredible (at being average). *Ethics and Information Technology*, 27(3):36.

William Rudman, Catherine Chen, and Carsten Eickhoff. 2023. Outlier dimensions encode task-specific knowledge. *arXiv preprint arXiv:2310.17715*.

Martin Ruef and Colin Birkhead. 2024. Learning from outliers and anomalies. *Academy of Management Perspectives*, (ja):amp–2023.

Avgi Saketopoulou. 2023. *Sexuality Beyond Consent: Risk, Race, Traumatophilia*. NYU Press.

Rylan Schaeffer, Joshua Kazdan, Alvan Caleb Arulandu, and Sanmi Koyejo. 2025. Position: Model collapse does not mean what you think. *Preprint*, arXiv:2503.03150.

Reva Schwartz, Reva Schwartz, Apostol Vassilev, Kristen Greene, Lori Perine, Andrew Burt, and Patrick Hall. 2022. *Towards a standard for identifying and managing bias in artificial intelligence*, volume 3. US Department of Commerce, National Institute of Standards and Technology ....

Andrew D Selbst, Danah Boyd, Sorelle A Friedler, Suresh Venkatasubramanian, and Janet Vertesi. 2019. Fairness and abstraction in sociotechnical systems. In *Proceedings of the conference on fairness, accountability, and transparency*, pages 59–68.

Chantal Shaib, Joe Barrow, Jiuding Sun, Alexa F Siu, Byron C Wallace, and Ani Nenkova. 2024a. Standardizing the measurement of text diversity: A tool and a comparative analysis of scores. *arXiv preprint arXiv:2403.00553*.

Chantal Shaib, Yanai Elazar, Junyi Jessy Li, and Byron C. Wallace. 2024b. Detection and measurement of syntactic templates in generated text. *Preprint*, arXiv:2407.00211.

Renee Shelby, Shalaleh Rismani, Kathryn Henne, AJung Moon, Negar Rostamzadeh, Paul Nicholas, N'Mah Yilla, Jess Gallegos, Andrew Smart, Emilio Garcia, and Gurleen Virk. 2023. Sociotechnical harms of algorithmic systems: Scoping a taxonomy for harm reduction. *Preprint*, arXiv:2210.05791.

Ivaxi Sheth, Jan Wehner, Sahar Abdelnabi, Ruta Binkyte, and Mario Fritz. 2025. Safety is essential for responsible open-ended systems. *Preprint*, arXiv:2502.04512.

Kacper Sokol and Eyke Hüllermeier. 2025. All you need for counterfactual explainability is principled and reliable estimate of aleatoric and epistemic uncertainty. *Preprint*, arXiv:2502.17007.

Yuda Song, Julia Kempe, and Remi Munos. 2025. Outcome-based exploration for llm reasoning. *Preprint*, arXiv:2509.06941.

Zhivar Sourati, Alireza S. Ziabari, and Morteza Dehghani. 2025. The homogenizing effect of large language models on human expression and thought. *Preprint*, arXiv:2508.01491.

Gayatri Chakravorty Spivak. 1988. Can the subaltern speak?

Kenneth O. Stanley and Joel Lehman. 2015. *Why Greatness Cannot Be Planned: The Myth of the Objective*. Springer Cham.

Peiqi Sui, Eamon Duede, Sophie Wu, and Richard Jean So. 2024. Confabulation: The surprising value of large language model hallucinations. *Preprint*, arXiv:2406.04175.

Guangyan Sun, Mingyu Jin, Zhenting Wang, James Chenhao Liang, Tong Geng, Qiang Guan, Qifan Wang, Mengnan Du, Yongfeng Zhang, Dongfang Liu, and 1 others. 2025. Hallucinating llm could be creative.

Koray Tahiroglu and Lonce Wyse. 2024. Latent spaces as platforms for sonic creativity. In *Proceedings of the 16th International Conference on Computational Creativity, ICCC*, volume 24.

Nassim N Taleb. 2013. 'antifragility' as a mathematical idea. *Nature*, 494(7438):430–430.

Richard G. Tedeschi and Lawrence G. Calhoun. 1996. The posttraumatic growth inventory: Measuring the positive legacy of trauma. *Journal of Traumatic Stress*, 9(3):455–471.

Guy Tevet and Jonathan Berant. 2021. Evaluating the evaluation of diversity in natural language generation. In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Main Volume*, pages 326–346.

Hoang Thanh-Tung and Truyen Tran. 2020. Catastrophic forgetting and mode collapse in gans. In *2020 international joint conference on neural networks (ijcnn)*, pages 1–10. IEEE.

Brian Uzzi, Satyam Mukherjee, Michael Stringer, and Ben Jones. 2013. Atypical combinations and scientific impact. *Science*, 342(6157):468–472.

Sahil Verma and Julia Rubin. 2018. Fairness definitions explained. In *Proceedings of the 2018 IEEE/ACM International Workshop on Software Fairness (FairWare)*, pages 1–7. IEEE.

Eric Von Hippel. 1989. New product ideas from 'lead users'. *Research-Technology Management*, 32(3):24–27.

Sanne Vrijenhoek, Savvina Daniil, Jorden Sandel, and Laura Hollink. 2024. Diversity of what? on the different conceptualizations of diversity in recommender systems. In *The 2024 ACM Conference on Fairness Accountability and Transparency*, FAccT '24, page 573–584. ACM.

Sandra Wachter, Brent Mittelstadt, and Chris Russell. 2021. Why fairness cannot be automated: Bridging the gap between eu non-discrimination law and ai. *Computer Law & Security Review*, 41:105567.

McKenzie Wark. 2023. *Raving*. Practices. Duke University Press, Durham, NC.

Peter West and Christopher Potts. 2025. Base models beat aligned models at randomness and creativity. *arXiv preprint arXiv:2505.00047*.

James Woodward. 2005. *Making things happen: A theory of causal explanation*. Oxford university press.

Lingfei Wu, Dashun Wang, and James A Evans. 2019. Large teams develop and small teams disrupt science and technology. *Nature*, 566(7744):378–382.

Chenghao Yang and Ari Holtzman. 2025. Llm probability concentration: How alignment shrinks the generative horizon. *Preprint*, arXiv:2506.17871.

Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Thomas L. Griffiths, Yuan Cao, and Karthik Narasimhan. 2023. Tree of thoughts: Deliberate problem solving with large language models. *Preprint*, arXiv:2305.10601.

Yasin Yazici, Chuan-Sheng Foo, Stefan Winkler, Kim-Hui Yap, and Vijay Chandrasekhar. 2020. Empirical analysis of overfitting and mode drop in gan training. *Preprint*, arXiv:2006.14265.

Shuzhou Yuan, Zhan Qu, Ashish Yashwanth Kangen, and Michael Färber. 2025. Can hallucinations help? boosting llms for drug discovery. *Preprint*, arXiv:2501.13824.

Jiayi Zhang, Simon Yu, Derek Chong, Anthony Sicilia, Michael R. Tomz, Christopher D. Manning, and Weiyan Shi. 2025a. Verbalized sampling: How to mitigate mode collapse and unlock llm diversity. *Preprint*, arXiv:2510.01171.

Yiming Zhang, Harshita Diddee, Susan Holm, Hanchen Liu, Xinyue Liu, Vinay Samuel, Barry Wang, and Daphne Ippolito. 2025b. Noveltybench: Evaluating language models for humanlike diversity. *Preprint*, arXiv:2504.05228.

Lianghui Zhu, Xinggang Wang, and Xinlong Wang. 2025. Judgelm: Fine-tuned large language models are scalable judges. In *International Conference on Learning Representations (ICLR)*.

Yaoming Zhu, Sidi Lu, Lei Zheng, Jiaxian Guo, Weinan Zhang, Jun Wang, and Yong Yu. 2018. Texygen: A benchmarking platform for text generation models. *Preprint*, arXiv:1802.01886.

P. Zurn, A.J. Pitts, T.M. Bettcher, and PJ DiPietro. 2024. *Trans Philosophy*. University of Minnesota Press.

A Implementing generalized diversities

Our structure-aware language is intentionally abstract so it **admits multiple implementations**, not only the one we presented in the main paper. In this appendix, we think through two alternative choices:

1. Generalization of the structure core through the escort power mean
2. Reinterpretation of the deviance as relative entropy

Our goal with this appendix is to **inspire reflection** on diversity beyond what was explicitly presented in our framework.

A.1 Generalizing the structure core

Inspired by value measures (Leinster, 2024) and escort distributions (Bercher, 2011), we generalize the structure core as the **escort power mean**:

$$\langle \alpha_{i(q,r)} \rangle(x_p) = \left(\frac{\sum_{y \in \mathsf{Str}_\top(x_p)} p(y|x_p)^r \alpha_i(y)^q}{\sum_{y \in \mathsf{Str}_\top(x_p)} p(y|x_p)^r} \right)^{1/q} \tag{A.1}$$

We simplify the notation by introducing the escort distribution:

$$p_{(r)}(y|x_p) = \frac{p(y|x_p)^r}{\sum_{y \in \mathsf{Str}_\top(x_p)} p(y|x_p)^r} \tag{A.2}$$

Then, the **generalized structure core** is written as:

$$\langle \alpha_{i(q,r)} \rangle(x_p) = \left(\mathbb{E}_{y \sim p_{(r)}(\cdot|x_p)} [\alpha_i(y)^q] \right)^{1/q} \tag{A.3}$$

When $q = 1$ and $r = 1$, the generalized structure core recovers our original structure core in Equation 5 and Equation 9. Different values for q, r give us alternative interesting cores. For instance:

$$\begin{aligned}
\langle \alpha_{i(1,0)} \rangle(x_p) &= \frac{1}{|\mathsf{Str}_\top(x_p)|} \sum_{y \in \mathsf{Str}_\top(x_p)} \alpha_i(y) \\
\langle \alpha_{i(1,\infty)} \rangle(x_p) &= \alpha_i(\arg\max_y p(y|x_p)) \\
\langle \alpha_{i(\infty,1)} \rangle(x_p) &= \max_{y \in \mathrm{supp}(p(\cdot|x_p))} \alpha_i(y) \\
\langle \alpha_{i(-\infty,\infty)} \rangle(x_p) &= \min_{y \in \mathrm{modes}(p(\cdot|x_p))} \alpha_i(y)
\end{aligned}$$

For a given structure α_i, we can think of q selecting whether large or small compliance values dominate, and r selecting whether the large body or long-tails of $p(\cdot|x_p)$ dominate. **By parameterizing, we make transparent how we weigh rarity, signal strength, and balance**. Since different parameters reflect different viewpoints (Leinster, 2024), we shall always consider a full diversity profile before drawing conclusions about how our interventions impact diversity.

A.2 Reinterpreting deviance

We can think of a **generalized orientation** as:

$$\theta_{n,k}(y|x_p) = \mathsf{orient}\left(\Lambda_n(y), \langle \Lambda_n \rangle(x_p)\right) \tag{A.4}$$

with $\mathsf{orient} : [0,1]^n \times [0,1]^n \to [0,1]^k$.

Then, the **generalized deviance** is:

$$\begin{aligned}
\partial_{n,k}(y|x_p) &= \|\theta_{n,k}(y|x_p)\|_{\mathsf{orient}} \\
\|\cdot\|_{\mathsf{orient}} &: [0,1]^k \to \mathbb{R}^+
\end{aligned} \tag{A.5}$$

If we choose $\mathsf{orient}(\Lambda_x, \Lambda_y) = \Lambda_x - \Lambda_y$ and $\|\cdot\|_{\mathsf{orient}} = \|\cdot\|_\theta$, we recover our original deviance in Equation 8 and Equation 9.

For **relative entropy**, we consider the **Rényi entropy** defined (Leinster, 2024) as:

$$H_q(\mathbf{p} \,\|\, \mathbf{r}) = \frac{1}{q-1} \log \sum_{i \in \mathrm{supp}(\mathbf{p})} p_i^q \, r_i^{1-q} \tag{A.6}$$

Then, we can think of a dummy $\mathsf{orient}()$ that just stores Λ_x, Λ_y and a $\|\cdot\|_{\mathsf{orient}}$ operator that computes the relative entropy between them. For a given normalized core $\langle \overline{\Lambda}_{\mathrm{norm}_n} \rangle = \{\langle \overline{\alpha}_{\mathrm{norm}_i} \rangle, \ldots\}$ and normalized system $\overline{\Lambda}_{\mathrm{norm}_n} = \{\overline{\alpha}_{\mathrm{norm}_i}, \ldots\}$, we define two Hill number (Leinster, 2024) deviances: the **excess deviance** and **deficit deviance**:

$$\partial_q^+(y, x_p) = e^{H_q(\overline{\Lambda}_{\mathrm{norm}_n}(y) \,\|\, \langle \overline{\Lambda}_{\mathrm{norm}_n} \rangle(x_p))} \tag{A.7}$$

$$\partial_q^-(y, x_p) = e^{H_q(\langle \overline{\Lambda}_{\mathrm{norm}_n} \rangle(x_p) \,\|\, \overline{\Lambda}_{\mathrm{norm}_n}(y))} \tag{A.8}$$

We could read ∂_q^+ as the effective **over-compliance** and ∂_q^- as the effective **under-compliance** with respect to the normative compliance.

For instance, as $q \to \infty$, we interpret:

- ∂_∞^+ as the largest excess of compliance

$$\partial_\infty^+ = \max_i \frac{\overline{\alpha}_{\mathrm{norm}_i}(y)}{\langle \overline{\alpha}_{\mathrm{norm}_i} \rangle(x_p)}$$

- ∂_∞^- as the largest deficit of compliance

$$\partial_\infty^- = \max_i \frac{\langle \overline{\alpha}_{\mathrm{norm}_i} \rangle(x_p)}{\overline{\alpha}_{\mathrm{norm}_i}(y)}$$

All of this to say, there are **multiple ways we can reason about structures and statistics jointly**. We encourage readers to develop alternative and competing formalisms that share our conceptual backbone: structures that make context explicit, cores that encode the normativity that homogenization pushes us toward, and orientations that capture perspectives of non-normativity. Above all, **we ask everyone to think deeper about diversity**.

B Theoretical touchpoints

In this appendix, we explore how our theoretical framework connects to other frameworks. To that purpose, we consider an unprompted scenario of a singleton system with binary compliance for its single structure:

$$\Lambda_*(x) := (\alpha_*(x)) \qquad \alpha_*(x) \in \{0, 1\}$$

Then, the structure core represents the probability of compliance being exactly 1:

$$\mu := \langle \alpha_* \rangle = \sum_{c \in \{0,1\}} c \, \Pr(\alpha = c) = \Pr(\alpha = 1)$$

Our singleton deviance is expressed as:

$$\partial_*(x) = \|\alpha_*(x) - \mu\|_\theta$$

B.1 Expected deviance and Gini-Simpson index

To calculate the expected deviance, we consider two choices for $\|\cdot\|_\theta$: absolute value and the squared ℓ_2 norm. For each, we find connections between $\mathbb{E}[\partial_*]$ and the Gini-Simpson index for a binary variable:

$$\mathbb{E}[|\alpha_* - \mu|] = 2\mu(1-\mu) = \mathrm{GS}$$

$$\mathbb{E}[\|\alpha_* - \mu\|_2^2] = \mathrm{Var}[\alpha_*] = \mu(1-\mu) = \frac{\mathrm{GS}}{2}$$

If we interpret GS as the degree of mixing in outcomes, then increasing the expected deviance drives heterogeneity rather than concentration.

B.2 Is-It-Valid classification for Hallucinations

To reason about hallucinations, authors in (Kalai et al., 2025) partition the space of plausible outputs into disjoint sets of valid outputs V and errors E. In their framework, a model hallucinates when it cannot solve the binary discrimination problem "Is-It-Valid?" (IIV). Their framework can be interpreted through our structure-aware language:

$$\alpha_{\mathrm{IIV}}(x) = \mathbf{1}[x \in V]$$

We can connect their generative hallucination rate given by $\mathrm{err} = \Pr_{x \sim \hat{p}}[x \in E] = \hat{p}(E)$ to the system core of a singleton IIV system:

$$\langle \alpha_{\mathrm{IIV}} \rangle = 1 - \mathrm{err}$$

The paper (Kalai et al., 2025) points out that future work should "consider degrees of hallucination". Our structure-aware framework provides the language to reason about these desired **graded notions of hallucination**: We can score a string under multiple structures, with scores encoding real-valued nuance beyond the binary.

B.3 Language Generation in the Limit

Recent work (Kleinberg and Mullainathan, 2024; Kalavasis et al., 2025a) studies language generation where a generator G, given strings from an unknown target language K, must output strings that are both **novel** and **valid**. We can re-interpret some of their framework as a special case of our structure-aware formulation.

Given a language collection $\mathcal{L} = \{L_1, L_2, \ldots\}$, we can define membership structures with corresponding cores that represent the probability of generating a string valid for each corresponding language:

$$\alpha_{L_i}(x) = \mathbf{1}[x \in L_i] \qquad \langle \alpha_{L_i} \rangle = \Pr[y \in L_i]$$

The literature is currently (Kalavasis et al., 2025b) exploring the trade-offs between consistency and breadth. An LLM generates strings consistent with our target language K if:

$$\langle \alpha_K \rangle = 1 \qquad \text{when} \qquad \mathbb{E}[\partial_K]_{y \sim p_{\mathsf{LLM}}} \to 0$$

An LLM generation has breadth when all strings of our target language $K \in \mathcal{L}$ can be generated:

$$\forall\, y \in K : \; p_{\mathsf{LLM}}(y) > 0 \iff K \subseteq \mathrm{supp}(p_{\mathsf{LLM}})$$

Our structure-aware framework gives us insight that homogenization is relative to a system. Indeed, pushing for consistency shall not imply that we push for homogenization in every context. Generally, for $\Lambda_K \neq \Lambda_m$:

$$\mathbb{E}[\partial_K] \to 0 \quad \neq \quad \mathbb{E}[\partial_m] \to 0$$

Thinking explicitly through structures and systems allows us to formulate interesting questions (for instance, is $\Lambda_K = \Lambda_{\text{IIV}}$?) that will help us make connections between all these theoretical efforts. We present these touchpoints as **starting points for deeper exploration**.

C Comparing with linguistic metrics of diversity

In this appendix, we place our structure-aware framework in the context of existing diversity metrics. There are two main categories of metrics of linguistic diversity: intrinsic and extrinsic.

C.1 Intrinsic linguistic diversity

Intrinsic diversity refers to the types of variation within a generated language without external references. The literature accounts (Guo et al., 2025; Tevet and Berant, 2021) for intrinsic diversity in both form and content.

C.1.1 Form Diversity

On the one hand, we have **syntactic** diversity, which accounts for the variety in sentence patterns. These metrics involve identifying patterns in sentences and subsequently comparing them based on their occurrence across the generated language. Some of the methods include: transforming sentences into part-of-speech (POS) tag sequences and evaluating redundancy through compression (Shaib et al., 2024b); and parsing text into trees and mapping the resulting graphs into a vector space (Guo et al., 2024a) or measuring their distribution (del Prado Martin, 2024).

Our framework naturally includes syntactic metrics as we can define syntactic systems in which each structure encodes a syntactic pattern of interest:

$$\Lambda_{\text{syntax}} = (\alpha_{\text{POS Tag}}, \alpha_{\text{Noun Phrase}}, \dots) \quad \text{(C.1)}$$

On the other hand, we have **lexical** diversity, which accounts for the variety in vocabulary. Generally, these surface-level metrics measure repetition and reuse (Kendro et al., 2025; Shaib et al., 2024a). Common approaches include counting unique n-grams (Li et al., 2016; Estève et al., 2025), measuring their overlap (Zhu et al., 2018), and calculating their entropy (Estève et al., 2025).

Our framework can account for lexical metrics in an analogous way that it accounts for syntax metrics. For instance, we could define lexical systems in which each structure encodes a unique n-gram. However, this might not be very practical, as the number of possible n-grams grows exponentially with vocabulary size (Jurafsky and Martin, 2009).

$$\Lambda_{\text{lexicon}} = (\alpha_{1-\text{gram}}, \dots) \quad \text{(C.2)}$$

C.1.2 Content Diversity

To measure **semantic** diversity, sentences are transformed into embeddings that can be used to measure similarity (Guo et al., 2025). The literature describes various ways to exploit these similarities, from calculating the effective number of unique elements in the sample through the eigenvalues of the similarity matrix (Friedman and Dieng, 2023), to quantifying the divergence in the intermediate reasoning steps taken by LLMs to find a solution (Ju et al., 2025).

In our framework, we can construct semantic systems in which the compliance of each structure is a measure of similarity to an internal reference embedding [10]. For instance, consider the following system:

$$\Lambda_{\text{semantics}} = (\alpha_{v_1}, \dots) \quad \text{(C.3)}$$

where each structure computes the dot product between the embedding vector of the string x and the unit vector v_i

$$\alpha_{v_i}(x) = \text{abs}(\text{embed}(x) \cdot v_i) \quad \text{(C.4)}$$

In this example, comparing the system compliance of two different strings ($\Lambda_{\text{semantics}}(x_a)$ vs $\Lambda_{\text{semantics}}(x_b)$) affords more interpretable measures of similarity and difference since we can decompose the results per v_i internal reference.

C.2 Extrinsic linguistic diversity

Extrinsic diversity metrics focus on divergence between a target (LLM-generated language) and an **external reference**, which could be text samples or real human language distributions (Pillutla et al., 2023). Comparisons between target and reference use the same methods (syntactic, lexical, and semantic) as those used for intrinsic diversity.

Our framework provides a language for reasoning about the target and reference through multiple

[10] These internal reference vectors might be the principal components of the learned embedding space, known concept vectors of interest, embeddings of prototypical sentences, and so on.

lenses. For example, we could ask questions that compare the target and the reference, like:

Are the same syntactic patterns present on average?

$$\|\langle\Lambda^{target}_{\text{syntax}}\rangle - \langle\Lambda^{reference}_{\text{syntax}}\rangle\|_\theta$$

Do they have the same range of semantic variety?

$$\mathbb{E}[\partial^{target}_{\text{semantics}}] \quad vs. \quad \mathbb{E}[\partial^{reference}_{\text{semantics}}]$$

Is toxic language equally likely after prefacing a text with "Be brutally honest."?

$$\langle\alpha^{target}_{\text{toxic}}\rangle(x_p) \quad vs. \quad \langle\alpha^{reference}_{\text{toxic}}\rangle(x_p)$$

with $x_p =$ "Be brutally honest."

Do they have the same ratio of syntactic and lexical diversity?

$$H(\langle\Lambda^{target}_{\text{Form}}\rangle) \quad vs. \quad H(\langle\Lambda^{reference}_{\text{Form}}\rangle)$$

with $\Lambda_{\text{Form}} = [\Lambda_{\text{syntax}}, \Lambda_{\text{lexicon}}]$

D Dynamics of relative diversity

As noted in Section 3.6, what is non-normative is conditional on what came before.[11] A prompt shifts the frame of reference. As a string is completed, the set of possible trajectories narrows, and so the system core and orientations change. Trajectories that were unreachable from the root of the tree may emerge as attractors once we condition on a specific subtree.

Tracking how cores and orientations evolve could reveal where diversity is most at stake during generation. Certain words in a sentence may act as branching points where the dynamics bifurcate dramatically. Identifying these could enable real-time Chain-of-Thought monitoring (Korbak et al., 2025).

D.1 Normative orders

Our framework allows us to define preorders. For a fixed system, LLM and prompt, we can rank strings by how deviant they are, and rank structures by how often strings comply with them:

$$\begin{aligned} x_a \preceq_{\partial_n} x_b \quad &\iff \quad \partial_n(x_a) \leq \partial_n(x_b) \\ \alpha_i \preceq_{\langle\cdot\rangle} \alpha_j \quad &\iff \quad \langle\alpha_i\rangle \leq \langle\alpha_j\rangle \end{aligned} \tag{D.1}$$

These orderings are not fixed. They depend on the prompt. A string that is deviant in one context may be normative in another. A structure that dominates early in generation may become rare as the trajectory unfolds.

[11] While homogenization reproduces "the same" and narrows futurability (Berardi, 2017), xeno-reproduction reproduces "the strange" and widens possibilities.

D.2 States and dynamics

Given a trajectory $y = x_T$, for $k \in \{0, 1, \ldots, T\}$, we can define **states** for all the intermediate continuations:

$$\phi^{(x)}_k = \langle\Lambda_n\rangle(x_k) \quad \phi^{(y)}_k = \theta_n(x_k|x_0) \quad \phi^{(z)}_k = \theta_n(y|x_k) \tag{D.2}$$

which form a discrete-time **dynamics**:

$$(\phi^{(x)}_0, \phi^{(y)}_0, \phi^{(z)}_0) \to \ldots \to (\phi^{(x)}_T, \phi^{(y)}_T, \phi^{(z)}_T)$$

D.3 Interpreting the states

The state $\phi^{(x)}$ evolves from the expected system compliance of all possible continuations at $\phi^{(x)}_0 = \langle\Lambda_n\rangle(\perp)$, to the specific system compliance of the trajectory at $\phi^{(x)}_T = \langle\Lambda_n\rangle(y) = \Lambda_n(y)$.

The state $\phi^{(y)}$ encodes how much the current path has deviated from normativity. It evolves from $\phi^{(y)}_0 = \theta_n(\perp|\perp) = \Lambda_n(\perp) - \langle\Lambda_n\rangle(\perp)$ to the full trajectory's orientation in the largest frame of reference at $\phi^{(y)}_T = \theta_n(y|\perp) = \Lambda_n(y) - \langle\Lambda_n\rangle(\perp)$.

The state $\phi^{(z)}$ represents how deviant the trajectory is in the largest frame of reference. It evolves from $\phi^{(z)}_0 = \theta_n(y|\perp) = \phi^{(y)}_T = \Lambda_n(y) - \langle\Lambda_n\rangle(\perp)$, to zero deviance[12] at $\phi^{(z)}_T = \theta_n(y|y) = 0$.

E Trajectory-level formulation

We present two complementary formulations of xeno-reproduction. The distribution-level formulation (Section 5) accounts for how interventions shape the entire probability landscape. The trajectory-level formulation reinterprets distribution-level scores as reward signals for individual output trajectories. Both formulations share the same underlying values but differ in their computational affordances.

While the distribution-level formulation reasons about expected statistics over all possible trajectories, the trajectory-level approach enables sample-based approximations that are more tractable in practice.

We define three component rewards. The **deviance reward** directly measures how far a trajectory deviates from the core:

$$r_d(y|x_p) = \partial_n(y|x_p) \tag{E.1}$$

The **fairness reward** upweights compliance with underrepresented structures, where the

[12] A zero deviance is when an orientation has a deviation value of zero for all structures.

weight for each structure is inversely proportional to its core value:

$$r_f(y|x_p) = \sum_i^n \mathsf{v}_i \alpha_i(y) \qquad \mathsf{v}_i \propto (\langle \alpha_i \rangle (x_p))^{-1} \tag{E.2}$$

The **constraint reward** encodes explicit preferences about which structures to target, avoid, or conserve:

$$r_c(y|x_p) = \sum_{t \in \mathsf{target}} \alpha_t(y) - \sum_{a \in \mathsf{avoid}} \alpha_a(y) - \sum_{c \in \mathsf{conserve}} |\alpha_c(y) - \langle \alpha_c \rangle (x_p)| \tag{E.3}$$

The **stray reward** combines these components with tunable weights:

$$r_\chi(y|x_p) = \lambda_d r_d(y|x_p) + \lambda_f r_f(y|x_p) + \lambda_c r_c(y|x_p) \tag{E.4}$$

Finally, we formulate xeno-reproduction as a **search over trajectories**:

$$p(y|x_p, w) \propto p(y|x_p, w_0)\, e^{\beta_r r_\chi(y|x_p)} \tag{E.5}$$

where w_0 represents the base model and β_r is a tunable temperature parameter. This formulation connects to best-of-n sampling and reward-weighted regression approaches in reinforcement learning from human feedback (RLHF), suggesting natural implementation strategies for xeno-reproduction.